

HW2_manideep

2025-02-14

##Problem 1

```
#install.packages("faraway")
library(faraway)
data(teengamb)
#print(teengamb, n =10)
```

#Part a

```
model <- lm(gamble ~ sex + status + income + verbal, data = teengamb)
summary(model)
```

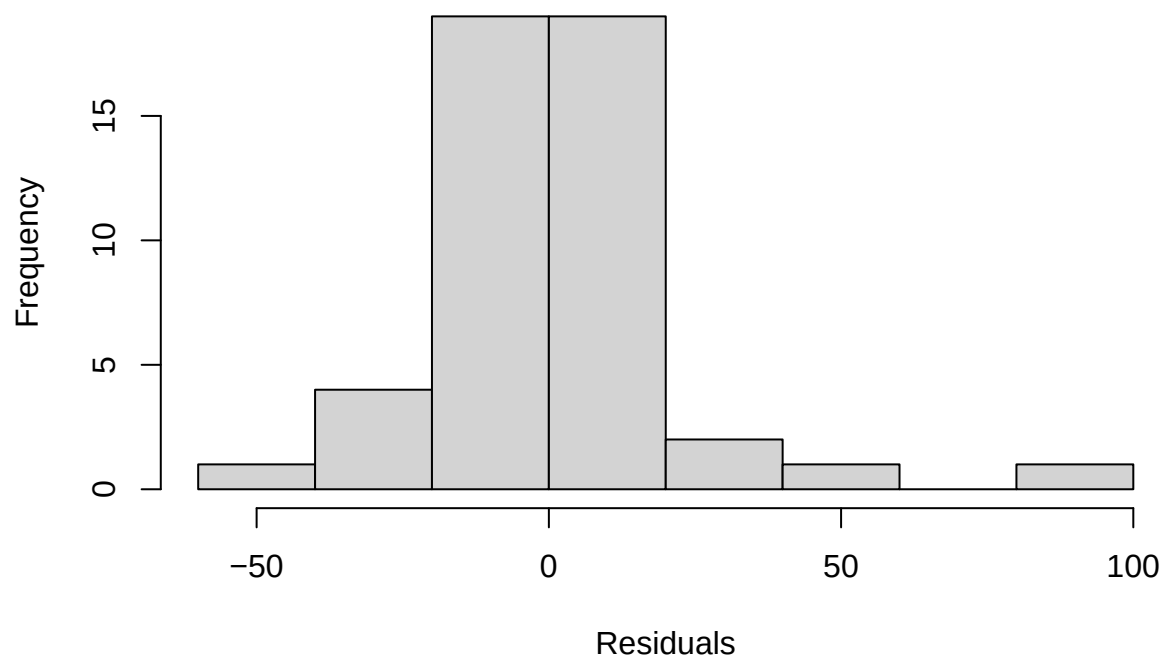
```
##
## Call:
## lm(formula = gamble ~ sex + status + income + verbal, data = teengamb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -51.082 -11.320  -1.451   9.452  94.252
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.55565   17.19680   1.312   0.1968
## sex         -22.11833    8.21111  -2.694   0.0101 *
## status        0.05223    0.28111   0.186   0.8535
## income        4.96198    1.02539   4.839 1.79e-05 ***
## verbal       -2.95949    2.17215  -1.362   0.1803
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.69 on 42 degrees of freedom
## Multiple R-squared:  0.5267, Adjusted R-squared:  0.4816
## F-statistic: 11.69 on 4 and 42 DF,  p-value: 1.815e-06
```

Here, the coefficient of sex is -22.11833, this means that males gamble 22.11833 pounds less than females per year. (note p-value is 0.0101 which is less than 0.05)

#Part b

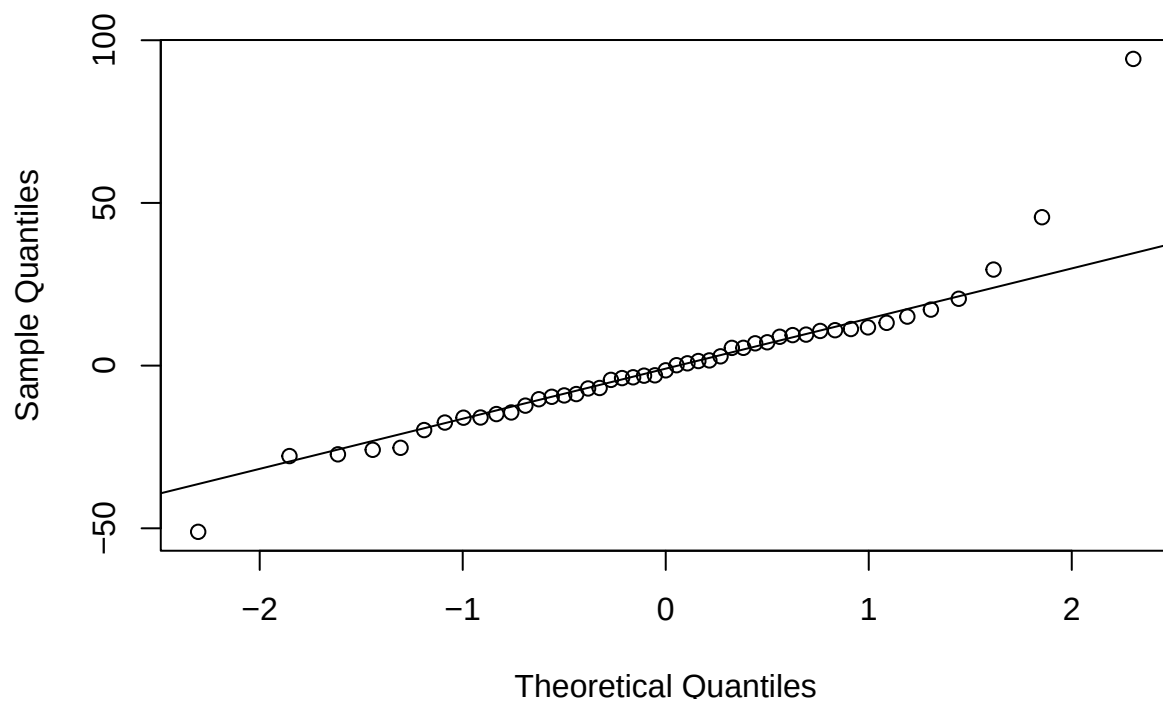
```
hist(residuals(model), main="Histogram of Residuals", xlab="Residuals")
```

Histogram of Residuals

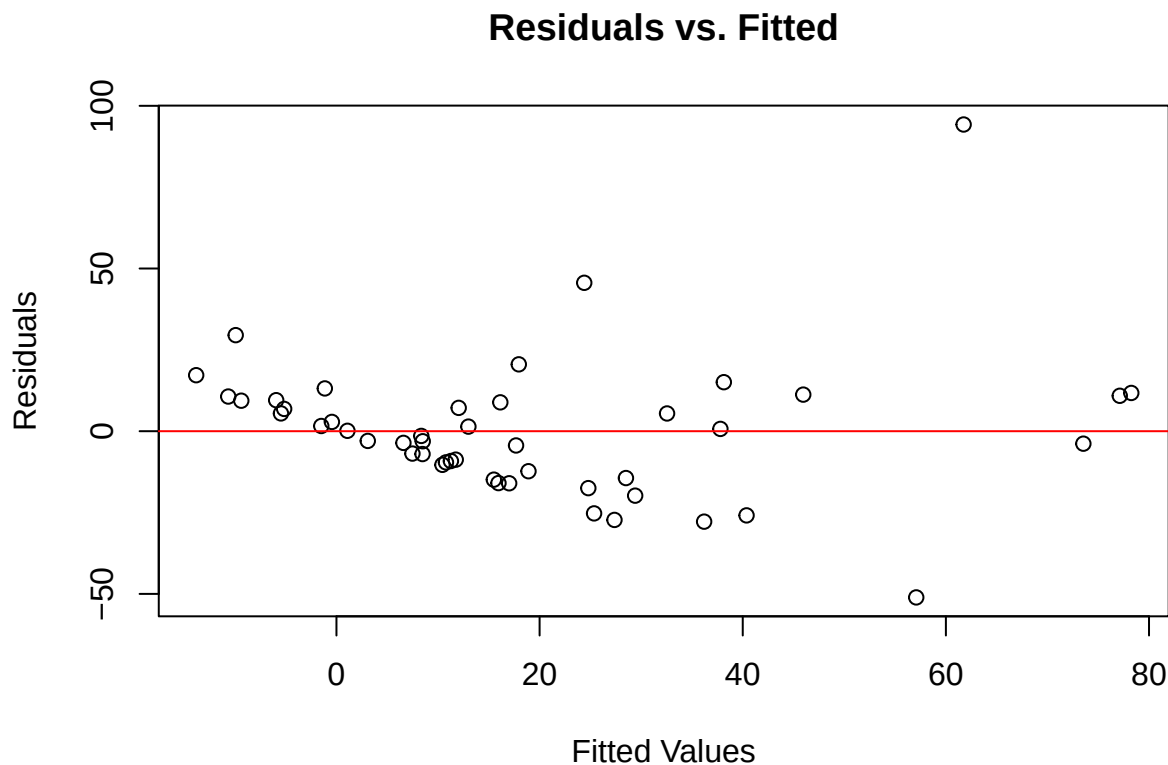


```
qqnorm(residuals(model))  
qqline(residuals(model))
```

Normal Q-Q Plot



```
plot(fitted(model), residuals(model),  
     main="Residuals vs. Fitted", xlab="Fitted Values", ylab="Residuals")  
abline(h=0, col="red")
```



his-
togram is not symmetric. residuals vs fitted plot shows that some points are spread out at certain values meaning non constant variance. normal Q-Q plot shows that at the end residuals are not normal. all of this confirm that the data is skewed. a log transformation would be a suitable option.

```
teengamb$log_gamble <- log(teengamb$gamble + 1)

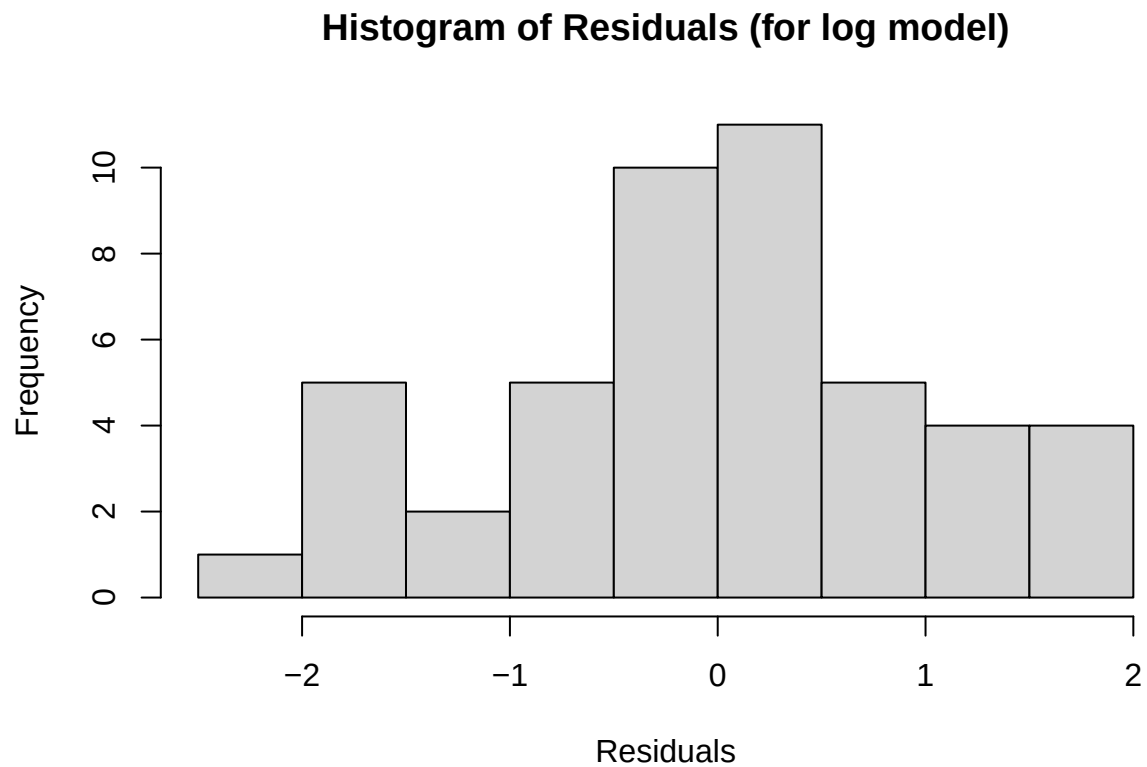
log_model <- lm(log_gamble ~ sex + status + income + verbal, data = teengamb)

summary(log_model)
```

```
##
## Call:
## lm(formula = log_gamble ~ sex + status + income + verbal, data = teengamb)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.35012 -0.56865  0.00413  0.71512  1.90319
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.71620    0.82240   2.087  0.0430 *
## sex           -0.87120    0.39268  -2.219  0.0320 *
## status         0.02983    0.01344   2.219  0.0320 *
## income         0.21565    0.04904   4.398 7.33e-05 ***
## verbal        -0.26165    0.10388  -2.519  0.0157 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.085 on 42 degrees of freedom
```

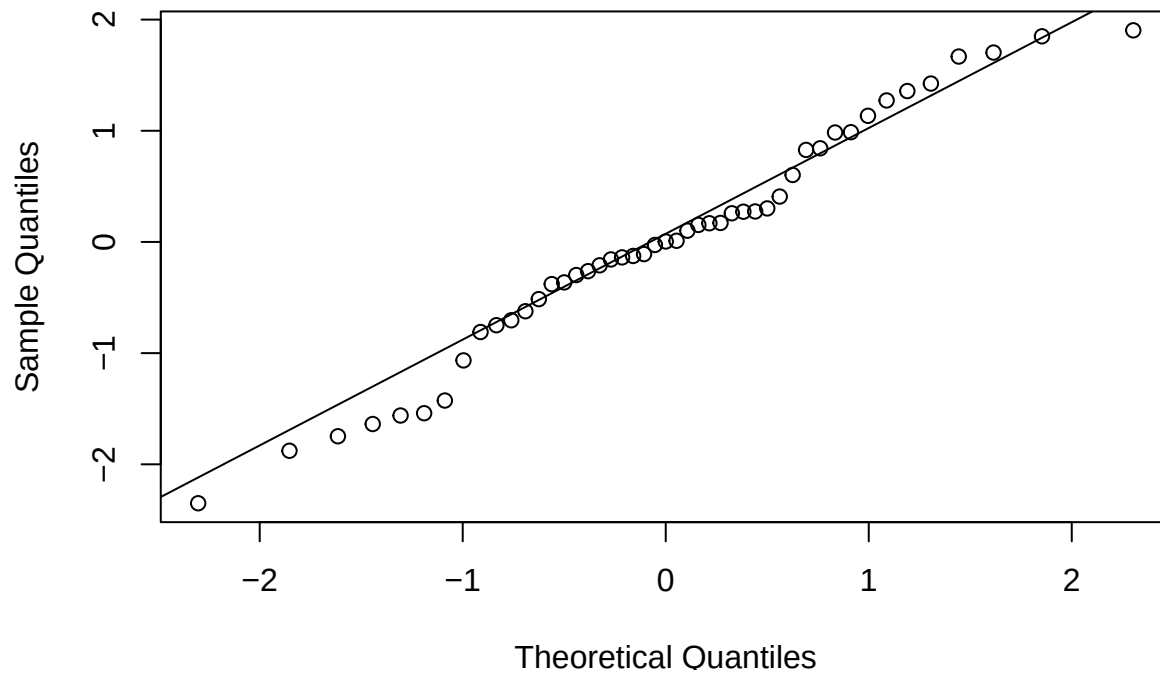
```
## Multiple R-squared:  0.5206, Adjusted R-squared:  0.475  
## F-statistic:  11.4 on 4 and 42 DF,  p-value: 2.347e-06
```

```
hist(residuals(log_model), main="Histogram of Residuals (for log model)", xlab="Residuals")
```



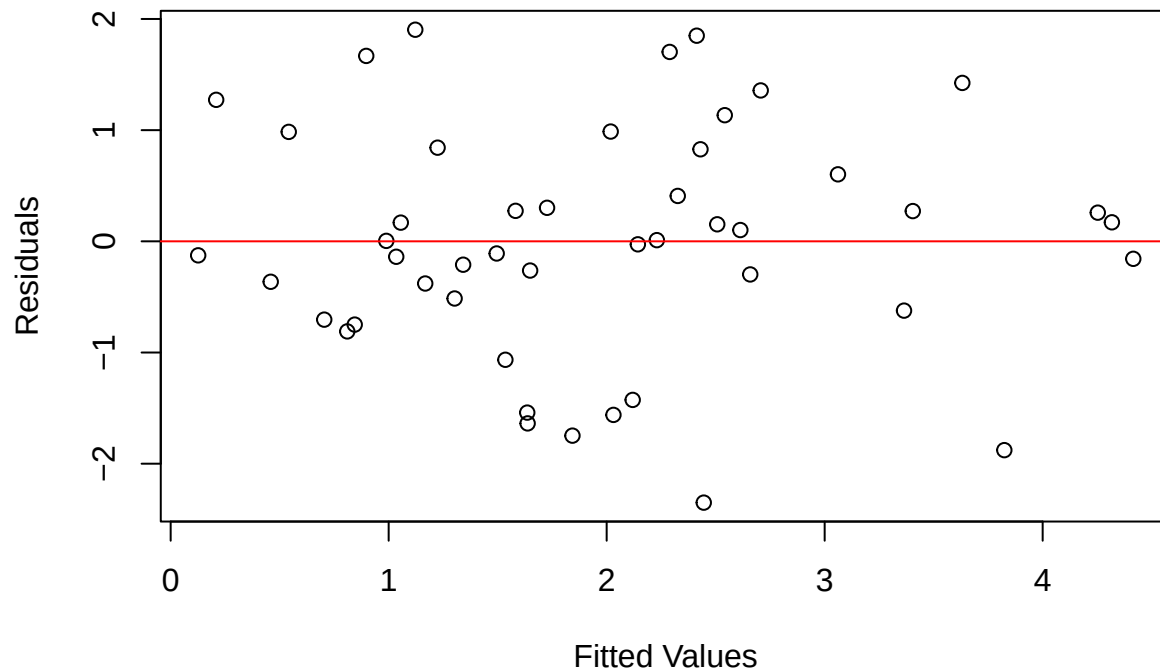
```
qqnorm(residuals(log_model), main = "Normal Q-Q Plot (Log Model)")  
qqline(residuals(log_model))
```

Normal Q-Q Plot (Log Model)



```
plot(fitted(log_model), residuals(log_model),  
     main="Residuals vs. Fitted (for log model) ", xlab="Fitted Values", ylab="Residuals")  
abline(h=0, col="red")
```

Residuals vs. Fitted (for log model)



Now, we can see that now the variance is now more constant, and the residuals are more normally distributed.

Part C

```
log_model_with_income <- lm(log_gamble ~ income, data = teengamb)
log_model <- lm(log_gamble ~ sex + status + income + verbal, data = teengamb)
anova(log_model_with_income, log_model)
```

```
## Analysis of Variance Table
##
## Model 1: log_gamble ~ income
## Model 2: log_gamble ~ sex + status + income + verbal
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      45 76.643
## 2      42 49.455  3    27.188 7.6966 0.0003293 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Here, H_0 is additional parameters donot improve the model, H_1 is additional parameters do improve the model. p-value is 0.0003293 (it is less than 0.01), so we reject null hypothesis, and we conclude additional parameters do improve the model.f # Part D

```
confint(log_model, "income", level = 0.95)
```

```
##           2.5 %      97.5 %
## income 0.1166847 0.3146076
```

We got 95% confidence interval for the income coefficient in the log-model is (0.1167, 0.3146). We can see that the effect of income is positive and significant. #Part E False, It is not what prediction interval means.95% prediction interval means that for a randomly selected teen who is male, has socioeconomic status score of 75, has income of 3 pounds per week, and a verbal score of 4 will fall between (0.585, 104.225) 95% percent of time.

#Problem 2

```
#install.packages("readxl")
library(readxl)
```

```
birth_rate <- read_excel("/Users/manideependyala/Downloads/birth_dat.xlsx")
print(birth_rate, n=10)
```

```
## # A tibble: 30 x 5
##   Nation      Birthrate PerCapIncome PopFarms InfantMort
##   <chr>         <dbl>         <dbl>     <dbl>     <dbl>
## 1 Venezuela      46.4           392       0.4       68.5
## 2 Mexico         45.7           118       0.61      87.8
## 3 Ecuador        45.3            44       0.53     116.
## 4 Colombia       38.6           158       0.53     107.
## 5 Ceylon         37.2            81       0.53     71.6
## 6 Puerto Rico    35             374       0.37     60.2
## 7 Chile          34             187       0.3     119.
## 8 Canada         28.3           993       0.19     33.7
## 9 United States  24.7          1723       0.12     27.2
## 10 Argentina     24.7           287       0.2       62
## # i 20 more rows
```

Part A

```
set.seed(123)
rand <- sample(1:nrow(birth_rate))
train_size <- round(2/3 * nrow(birth_rate))
train_data <- birth_rate[rand[1:train_size], ]
test_data <- birth_rate[rand[(train_size + 1):nrow(birth_rate)], ]
head(train_data)

## # A tibble: 6 x 5
##   Nation      Birthrate PerCapIncome PopFarms InfantMort
##   <chr>      <dbl>      <dbl>    <dbl>    <dbl>
## 1 Finland    21.6         688     0.34     32.4
## 2 Spain      20.3         244     0.48     56.5
## 3 Netherlands 21.7         575     0.14     21.6
## 4 Ecuador    45.3          44     0.53    116.
## 5 Argentina  24.7         287     0.2      62
## 6 Japan      20.8         239     0.42     46.7

head(test_data)

## # A tibble: 6 x 5
##   Nation      Birthrate PerCapIncome PopFarms InfantMort
##   <chr>      <dbl>      <dbl>    <dbl>    <dbl>
## 1 Colombia    38.6         158     0.53    107.
## 2 Venezuela   46.4         392     0.4      68.5
## 3 Italy        18          295     0.44     55.7
## 4 Greece      18.8         134     0.52     47.4
## 5 Sweden       15          910     0.24     18.7
## 6 Hungary     22.3         200     0.53     65.7

model <- lm(Birthrate ~ PerCapIncome + PopFarms + InfantMort, data = train_data)
summary(model)

##
## Call:
## lm(formula = Birthrate ~ PerCapIncome + PopFarms + InfantMort,
##     data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.228 -2.862 -1.292  3.514  7.555
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.643475   6.091058   0.270  0.79075
## PerCapIncome  0.007939   0.004497   1.765  0.09658 .
## PopFarms     21.073893  10.531752   2.001  0.06266 .
## InfantMort    0.230849   0.057811   3.993  0.00105 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.991 on 16 degrees of freedom
## Multiple R-squared:  0.6658, Adjusted R-squared:  0.6031
## F-statistic: 10.63 on 3 and 16 DF, p-value: 0.0004355
```

```
test_predictions <- predict(model, newdata = test_data)
test_mse <- sum((test_predictions - test_data$Birthrate)^2) / nrow(test_data)
print(paste("Test MSE:", round(test_mse, 5)))
```

```
## [1] "Test MSE: 109.43716"
```

This means on average the squared difference between predicted and actual birth rates is 109.43716. #Part B

```
set.seed(123)
mse_values <- numeric(10)

for (i in 1:10) {
  rand <- sample(1:nrow(birth_rate))
  train_size <- round(2/3 * nrow(birth_rate))
  train_data <- birth_rate[rand[1:train_size], ]
  test_data <- birth_rate[rand[(train_size + 1):nrow(birth_rate)], ]

  model <- lm(Birthrate ~ PerCapIncome + PopFarms + InfantMort, data = train_data)
  test_predictions <- predict(model, newdata = test_data)

  mse_values[i] <- sum((test_predictions - test_data$Birthrate)^2) / nrow(test_data)
}

mse_range <- range(mse_values)
mse_sd <- sd(mse_values)

print(mse_values)
```

```
## [1] 109.43716 20.16707 92.25910 82.52375 37.40118 94.23838 94.76744
## [8] 36.05029 46.33982 64.22363
```

```
print(paste("Range of test MSE:", mse_range[1], "-", mse_range[2]))
```

```
## [1] "Range of test MSE: 20.1670668643876 - 109.437161958342"
```

```
print(paste("Standard deviation of test MSE:", round(mse_sd, 5)))
```

```
## [1] "Standard deviation of test MSE: 31.00044"
```

#Part C

```
set.seed(123)
k <- 5
folds <- cut(seq(1, nrow(birth_rate)), breaks = k, labels = FALSE)
cross_validation_mse_values <- numeric(k)

for (i in 1:k) {
  test_indices <- which(folds == i, arr.ind = TRUE)
  train_data <- birth_rate[-test_indices, ]
  test_data <- birth_rate[test_indices, ]

  model <- lm(Birthrate ~ PerCapIncome + PopFarms + InfantMort, data = train_data)
  test_predictions <- predict(model, newdata = test_data)

  cross_validation_mse_values[i] <- sum((test_predictions - test_data$Birthrate)^2) / nrow(test_data)
}
```



```

cross_validation_mse_mean <- mean(cross_validation_mse_values)
cross_validation_mse_range <- range(cross_validation_mse_values)
cross_validation_mse_sd<- sd(cross_validation_mse_values)

print(cross_validation_mse_values)

## [1] 327.25372 56.96776 148.99818 34.61389 70.52200
print(paste("Mean Cross Validation MSE:", round(cross_validation_mse_mean, 5)))

## [1] "Mean Cross Validation MSE: 127.67111"
print(paste("Range of Cross Validation MSE:", cross_validation_mse_range[1], "-", cross_validation_mse_

## [1] "Range of Cross Validation MSE: 34.6138865140497 - 327.253720271698"
print(paste("Standard deviation of Cross Validation MSE:", round(cross_validation_mse_sd, 5)))

## [1] "Standard deviation of Cross Validation MSE: 119.59588"
#Part D
set.seed(123)
cross_validation_mse_repeats <- numeric(10)

for (j in 1:10) {
  folds <- cut(seq(1, nrow(birth_rate)), breaks = k, labels = FALSE)
  mse_values_cross_validation <- numeric(k)

  for (i in 1:k) {
    test_indices <- which(folds == i, arr.ind = TRUE)
    train_data <- birth_rate[-test_indices, ]
    test_data <- birth_rate[test_indices, ]

    model <- lm(Birthrate ~ PerCapIncome + PopFarms + InfantMort, data = train_data)
    test_predictions <- predict(model, newdata = test_data)

    mse_values_cross_validation[i] <- sum((test_predictions - test_data$Birthrate)^2) / nrow(test_data)
  }

  cross_validation_mse_repeats[j] <- mean(mse_values_cross_validation)
}

cross_validation_mse_range_10 <- range(cross_validation_mse_repeats)
cross_validation_mse_sd_10 <- sd(cross_validation_mse_repeats)

print(cross_validation_mse_repeats)

## [1] 127.6711 127.6711 127.6711 127.6711 127.6711 127.6711 127.6711 127.6711
## [9] 127.6711 127.6711
print(paste("Range of Cross Validation MSE (10 runs):", cross_validation_mse_range_10[1], "-", cross_va

## [1] "Range of Cross Validation MSE (10 runs): 127.671108419773 - 127.671108419773"
print(paste("Standard deviation of Cross Validation MSE (10 runs):", round(cross_validation_mse_sd_10, 1

## [1] "Standard deviation of Cross Validation MSE (10 runs): 0"

```

Validation set approach in part b, shows high variability with MSE values in (20.17,109.44) and a standard deviation of 31.00, indicating that results will fluctuate depending on the random split of training and test data. the 5-Fold Cross-Validation (Part d) gives a constant MSE of 127.67 across all 10 runs, with zero variability.

Question 3

Part A

$$\begin{aligned} E[(\hat{f}(x_0) - f(x_0))^2] &= \text{Var}(\hat{f}(x_0)) + \\ &\quad (E[\hat{f}(x_0)] - f(x_0))^2 \\ &= \text{Var}(\hat{f}(x_0)) + \text{Bias}_{f(x_0)}^2(\hat{f}(x_0)) \end{aligned}$$

Part B.

$$y = f(x_0) + \varepsilon_0$$

$$\begin{aligned} (y - \hat{f}(x_0))^2 &= (f(x) - \hat{f}(x_0) + \varepsilon_0)^2 \\ &= (f(x) - \hat{f}(x_0))^2 + \varepsilon_0^2 \\ &\quad + 2(f(x) - \hat{f}(x_0))\varepsilon_0 \end{aligned}$$

$$\begin{aligned} E[(y - \hat{f}(x_0))^2] &= E[(f(x) - \hat{f}(x_0))^2] + \\ &\quad E[\varepsilon_0^2] + 2E[(f(x) - \hat{f}(x_0))\varepsilon_0] \end{aligned}$$

since $E(\varepsilon_0) = 0$ and $f(x)$ and $\hat{f}(x_0)$ are independent, $E[(f(x) - \hat{f}(x_0))\varepsilon_0] = 0$

$$\text{W.L.T, } \text{Var}(W) = E[W^2] - (E[W])^2$$

$$\begin{aligned} \text{Var}(\varepsilon_0^2) &= E[\varepsilon_0^2] - (E[\varepsilon_0^2])^2 \\ &= E[\varepsilon_0^2] \end{aligned}$$

$$\therefore E[(y - \hat{f}(x_0))^2] = E[(f(x) - \hat{f}(x_0))^2] + E[\varepsilon_0^2]$$

$$E[(y - \hat{f}(x_0))^2] = E[(f(x) - \hat{f}(x_0))^2] + \text{Var}(\varepsilon_0)$$

$$= \text{Var}(\hat{f}(x_0)) + B_{f(x_0)}(\hat{f}(x_0))^2 + \text{Var}(\varepsilon_0)$$

(from part a)